

Modern Group Theory

An Interactive Problem-Centered Compendium



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Axioms and First Principles

This chapter introduces the fundamental structures of abstract algebra. The problems below automatically cycle through a 6-color palette to keep reading engaging.

1.1 Core Axioms

Definition 1.1: Group

A group $(G, \cdot)$ is a set G equipped with a binary operation satisfying:

1. **Closure:** $\forall a, b \in G, a \cdot b \in G$.
2. **Associativity:** $\forall a, b, c \in G, (a \cdot b) \cdot c = a \cdot (b \cdot c)$.
3. **Identity:** $\exists e \in G$ such that $\forall a \in G, e \cdot a = a \cdot e = a$.
4. **Inverse:** $\forall a \in G, \exists a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = e$.

Problem 1.1: Involution and Commutativity

Let G be a group where $x^2 = e$ for every $x \in G$. Prove that G must be Abelian.

Problem 1.2: One-Sided Identity and Inverse

Let G be a non-empty set with an associative binary operation $\cdot$. Assume the following:

1. There exists $e \in G$ such that for all $a \in G, e \cdot a = a$. (Left Identity)
2. For each $a \in G$ there exists $b \in G$ such that $b \cdot a = e$. (Left Inverse)

Prove that G is a group with the given operation $\cdot$.

Problem 1.3: Different Sided Inverse

Let G be a non-empty set with an associative binary operation $\cdot$. Assume the following:

1. There exists $e \in G$ such that for all $a \in G, e \cdot a = a$. (Left Identity)
2. For each $a \in G$ there exists $b \in G$ such that $a \cdot b = e$. (Left Inverse)

Give an one counterexample such that G is not group with the given operation $\cdot$.

Problem 1.4: Even Powers

Let G be a finite group of even order. Show that G contains at least one element $g \neq e$ such that $g^2 = e$.

Problem 1.5: Exponential

Let G be a finite Abelian group of cardinality n . Show that for each $g \in G$, $g^n = e$.

Problem 1.6

Let G be a finite group. Prove that for each $g \in G$ there exists m such that $g^m = e$.

2

Subgroups and Centers

2.1 Subgroups, Order, and Cyclic Groups

Definition 2.1: Subgroup

Let G be a group and H be a non-empty subset of G . If H forms a group under the same binary operation as G , then H is called a subgroup of G . We denote this by $H \leq G$.

Definition 2.2: Order of a Group and Element

The **order of a group** G , denoted by $|G|$, is the number of elements in the set G . The **order of an element** $g \in G$, denoted by $|g|$, is the smallest positive integer n such that $g^n = e$. If no such positive integer exists, g is said to have infinite order.

Definition 2.3: Cyclic Group

A group G is called **cyclic** if there exists an element $g \in G$ such that every element in G can be written as g^k for some integer k . Such an element g is called a generator of G , and we write $G = \langle g \rangle$.

Problem 2.1: Union of Two Subgroups

Prove that G cannot be union of two proper subgroup.

Problem 2.2: Generated Subgroup

Let S be a subgroup of G such that $S \neq G$. Prove that $\langle S^c \rangle = G$.

Problem 2.3: $SS = S$

If S is finite non-empty subset of G with $SS \subseteq S$. Show that S is a subgroup. Is it true if S is infinite?

Problem 2.4: Number of subgroups

A group that has only a finite number of subgroups must be finite.

Problem 2.5: Existence of an element

Let G be an abelian group such that there exists $a, b \in G$ such that $|a| = m$ and $|b| = n$ with $\gcd(m, n) = 1$. Show the existence of an element $g \in G$ such that $|g| = mn$.

Problem 2.6

Let G be an abelian group such that there exists $a, b \in G$ such that $|a| = m$ and $|b| = n$. Show the existence of an element $g \in G$ such that $|g| = mn$. [Use the previous problem]

Problem 2.7

Prove that subgroup of cyclic groups is cyclic subgroup.

Problem 2.8: Cyclic groups

Let G be a cyclic group of order n . Let d divides n , show that total number of elements in G of order d is $\phi(d)$.

Problem 2.9

Let G be a finite group in which the number of solutions in G of the equation $x^n = e$ at most n for each n . Prove that G must be a cyclic group.

Problem 2.10

If every subgroup of G is an abelian group, does it imply G is an abelian group?